DBMS LAB DAY-6

LAB ASSIGNMENT–6

EMP Table Structure and Records

Question 1. Create a table EMP with the given structure and insert the supplied employee records.

Column	Data type	Constraint
EMP_ID	NUMBER(4)	PRIMARY KEY, NOT NULL
FIRST_NAME	VARCHAR2(20)	NOT NULL
LAST_NAME	VARCHAR2(25)	NOT NULL
HIRE_DATE	DATE	NOT NULL
JOB_ID	VARCHAR2(10)	NOT NULL
SAL	NUMBER(8,2)	NOT NULL
DEPT_ID	NUMBER(4)	CHECK (DEPT_ID >= 10)

```
CREATE TABLE EMP (
    EMP_ID     NUMBER(4) CONSTRAINT EMP_PK PRIMARY KEY,
    FIRST_NAME VARCHAR2(20) NOT NULL,
    LAST_NAME  VARCHAR2(25) NOT NULL,
    HIRE_DATE  DATE NOT NULL,
    JOB_ID     VARCHAR2(10) NOT NULL,
    SAL        NUMBER(8,2) NOT NULL,
    DEPT_ID    NUMBER(4) CONSTRAINT EMP_DEPT_CK CHECK (DEPT_ID >= 10)
);
```

Output: Table created.

Insert the 11 employee records:

```
INSERT INTO EMP VALUES (198, 'Donald', 'Connell', TO_DATE('21-JUN-99','DD-MON-RR'), 'SH_CLERK', 2600, 50);
INSERT INTO EMP VALUES (199, 'Douglas', 'Grant', TO_DATE('13-JAN-98','DD-MON-RR'), 'SH_CLERK', 3000, 50);
INSERT INTO EMP VALUES (200, 'Jennifer', 'Whalen', TO_DATE('17-SEP-87','DD-MON-RR'), 'AD_ASST', 4400, 10);
INSERT INTO EMP VALUES (201, 'Michael', 'Hartstein', TO_DATE('19-JAN-99','DD-MON-RR'), 'IT_PROG', 6000, 20);
INSERT INTO EMP VALUES (202, 'Pat', 'Fay', TO_DATE('25-OCT-89','DD-MON-RR'), 'AC_MGR', 6500, 20);
INSERT INTO EMP VALUES (203, 'Susan', 'Mavris', TO_DATE('26-NOV-76','DD-MON-RR'), 'AD_VP', 7500, 40);
INSERT INTO EMP VALUES (204, 'Hermann', 'Baer', TO_DATE('23-AUG-95','DD-MON-RR'), 'AD_PRES', 9500, 90);
INSERT INTO EMP VALUES (205, 'Shelley', 'Higgins', TO_DATE('24-FEB-98','DD-MON-RR'), 'AC_MGR', 2300, 60);
INSERT INTO EMP VALUES (206, 'William', 'Gitz', TO_DATE('12-MAR-01','DD-MON-RR'), 'IT_PROG', 5000, 60);
INSERT INTO EMP VALUES (100, 'Steven', 'King', TO_DATE('15-JUN-02','DD-MON-RR'), 'AD_ASST', 8956, 100);
INSERT INTO EMP VALUES (101, 'Neena', 'Kochar', TO_DATE('10-JUL-03','DD-MON-RR'), 'SH_CLERK', 3400, 30);
```

Output: 11 rows inserted.

EMP table data:

EMP_ID	FIRST_NAME	LAST_NAME	HIRE_DATE	JOB_ID	SAL	DEPT_ID
198	Donald	Connell	21-Jun-99	SH_CLERK	2600	50
199	Douglas	Grant	13-Jan-98	SH_CLERK	3000	50
200	Jennifer	Whalen	17-Sep-87	AD_ASST	4400	10
201	Michael	Hartstein	19-Jan-99	IT_PROG	6000	20
202	Pat	Fay	25-Oct-89	AC_MGR	6500	20
203	Susan	Mavris	26-Nov-76	AD_VP	7500	40
204	Hermann	Baer	23-Aug-95	AD_PRES	9500	90
205	Shelley	Higgins	24-Feb-98	AC_MGR	2300	60
206	William	Gitz	12-Mar-01	IT_PROG	5000	60
100	Steven	King	15-Jun-02	AD_ASST	8956	100
101	Neena	Kochar	10-Jul-03	SH_CLERK	3400	30

2. Display the first name of all employees in ascending order. Give an appropriate alias name to the column.

SQL Query:

```
SELECT FIRST_NAME AS EMPLOYEE_NAME FROM EMP ORDER BY FIRST_NAME ASC;
```

Output:

EMPLOYEE_NAME
Donald
Douglas
Hermann
Jennifer
Michael
Neena
Pat
Shelley
Steven
Susan
William

3. Display the first name of all employees in descending order. Give an appropriate alias name to the column.

SQL Query:

```
SELECT FIRST_NAME AS EMPLOYEE_NAME FROM EMP ORDER BY FIRST_NAME DESC;
```

Output:

EMPLOYEE_NAME
William
Susan
Steven
Shelley
Pat
Neena
Michael
Jennifer
Hermann
Douglas
Donald

4. Display the hire date of all employees in ascending order.

SQL Query:

```
SELECT HIRE_DATE FROM EMP ORDER BY HIRE_DATE ASC;
```

Output:

HIRE_DATE
26-Nov-76
17-Sep-87
25-Oct-89
23-Aug-95
13-Jan-98
24-Feb-98
19-Jan-99
21-Jun-99
12-Mar-01
15-Jun-02
10-Jul-03

5. Display employee details whose first name starts with either J or M. Sort the result by employee last name.

SQL Query:

```
SELECT * FROM EMP WHERE FIRST_NAME LIKE 'J%' OR FIRST_NAME LIKE 'M%' ORDER BY LAST_NAME;
```

Output:

EMP_ID	FIRST_NAME	LAST_NAME	HIRE_DATE	JOB_ID	SAL	DEPT_ID
201	Michael	Hartstein	19-Jan-99	IT_PROG	6000	20
200	Jennifer	Whalen	17-Sep-87	AD_ASST	4400	10

6. Find the highest, lowest, average and sum of salary of all employees. Give aliases Max, Min, Avg and Sum respectively.

SQL Query:

```
SELECT MAX(SAL) AS "Max", MIN(SAL) AS "Min", ROUND(AVG(SAL),2) AS "Avg", SUM(SAL) AS "Sum" FROM EMP;
```

Output:

Max	Min	Avg	Sum
9500	2300	5377.82	59156

7. Find the highest, lowest, average and sum of salary of all employees of each individual job type.

SQL Query:

```
SELECT JOB_ID, MAX(SAL) AS MAX_SAL, MIN(SAL) AS MIN_SAL, ROUND(AVG(SAL),2) AS AVG_SAL, SUM(SAL) AS SUM_SAL FROM EMP
GROUP BY JOB_ID ORDER BY JOB_ID;
```

Output:

JOB_ID	MAX_SAL	MIN_SAL	AVG_SAL	SUM_SAL
AC_MGR	6500	2300	4400.00	8800
AD_ASST	8956	4400	6678.00	13356
AD_PRES	9500	9500	9500.00	9500
AD_VP	7500	7500	7500.00	7500
IT_PROG	6000	5000	5500.00	11000
SH_CLERK	3400	2600	3000.00	9000

8. Display the number of people under each job.

SQL Query:

```
SELECT JOB_ID, COUNT(*) AS NUMBER_OF_PEOPLE FROM EMP GROUP BY JOB_ID ORDER BY JOB_ID;
```

Output:

JOB_ID	NUMBER_OF_PEOPLE
AC_MGR	2
AD_ASST	2
AD_PRES	1
AD_VP	1
IT_PROG	2
SH_CLERK	3

9. Display the number of managers in the company without listing their employee IDs or names.

SQL Query:

```
SELECT COUNT(*) AS NUMBER_OF_MANAGERS FROM EMP WHERE JOB_ID LIKE '%MGR';
```

In the supplied data, AC_MGR is the manager job code; two employees have this job ID.

Output:

NUMBER_OF_MANAGERS
2

10. Find the difference between the highest and the lowest salaries.

SQL Query:

```
SELECT MAX(SAL) - MIN(SAL) AS SALARY_DIFFERENCE FROM EMP;
```

Output:

SALARY_DIFFERENCE
7200

11. Display the maximum and average salary of the engineers.

SQL Query:

```
SELECT MAX(SAL) AS MAX_SALARY, ROUND(AVG(SAL),2) AS AVG_SALARY FROM EMP WHERE JOB_ID = 'IT_PROG';
```

IT_PROG is treated as the engineering designation in the supplied data.

Output:

MAX_SALARY	AVG_SALARY
6000	5500.00

12. Display the employee first name that is first and the employee first name that is last in the alphabetized list of all employees.

SQL Query:

```
SELECT MIN(FIRST_NAME) AS FIRST_NAME, MAX(FIRST_NAME) AS LAST_NAME_IN_ALPHABET FROM EMP;
```

Output:

FIRST_NAME	LAST_NAME_IN_ALPHABET
Douglas	William

13. Display the date when the first employee was hired and the last date of hiring. Rename the columns First Hire_Date and Last Hire_Date.

SQL Query:

```
SELECT MIN(HIRE_DATE) AS "First Hire_Date", MAX(HIRE_DATE) AS "Last Hire_Date" FROM EMP;
```

Output:

First Hire_Date	Last Hire_Date
26-Nov-76	10-Jul-03

14. Display the maximum and average salary of the clerks.

SQL Query:

```
SELECT MAX(SAL) AS MAX_SALARY, ROUND(AVG(SAL),2) AS AVG_SALARY FROM EMP WHERE JOB_ID = 'SH_CLERK';
```

Output:

MAX_SALARY	AVG_SALARY
3400	3000.00

15. Display the department number and the salary of the lowest-paid employee for each department. Give an alias to the minimum salary column.

SQL Query:

```
SELECT DEPT_ID, MIN(SAL) AS MIN_SALARY FROM EMP GROUP BY DEPT_ID ORDER BY DEPT_ID;
```

Output:

DEPT_ID	MIN_SALARY
10	4400
20	6000
30	3400
40	7500
50	2600
60	2300
90	9500
100	8956

16. Display the department number and the salary of the lowest-paid employee for each department. Exclude groups where the minimum salary is 3000 or less. Sort in descending order of salary.

SQL Query:

```
SELECT DEPT_ID, MIN(SAL) AS MIN_SALARY FROM EMP GROUP BY DEPT_ID HAVING MIN(SAL) > 3000 ORDER BY MIN(SAL) DESC;
```

Output:

DEPT_ID	MIN_SALARY
90	9500
100	8956
40	7500
20	6000
10	4400
30	3400

17. Display for all rows: first name, designation (job ID), current salary and three times the current salary as the desired salary.

SQL Query:

```
SELECT FIRST_NAME || ' whose designation is ' || JOB_ID || ' gets ' || SAL || ' but wants to earn ' || (3*SAL) AS
EMPLOYEE_SALARY_DETAILS FROM EMP ORDER BY EMP_ID;
```

Output:

EMPLOYEE_SALARY_DETAILS
Donald whose designation is SH_CLERK gets 2600 but wants to earn 7800
Douglas whose designation is SH_CLERK gets 3000 but wants to earn 9000
Jennifer whose designation is AD_ASST gets 4400 but wants to earn 13200
Michael whose designation is IT_PROG gets 6000 but wants to earn 18000
Pat whose designation is AC_MGR gets 6500 but wants to earn 19500
Susan whose designation is AD_VP gets 7500 but wants to earn 22500
Hermann whose designation is AD_PRES gets 9500 but wants to earn 28500
Shelley whose designation is AC_MGR gets 2300 but wants to earn 6900
William whose designation is IT_PROG gets 5000 but wants to earn 15000
Steven whose designation is AD_ASST gets 8956 but wants to earn 26868
Neena whose designation is SH_CLERK gets 3400 but wants to earn 10200

18. Display the details of all employees in the format shown in the question.

SQL Query:

```
SELECT EMP_ID || ', ' || FIRST_NAME || ', ' || LAST_NAME || ', ' || TO_CHAR(HIRE_DATE,'DD-MON-RR') || ', ' ||
JOB_ID || ', ' || TO_CHAR(SAL) || ', ' || TO_CHAR(DEPT_ID) AS EMPLOYEE_DETAILS FROM EMP ORDER BY EMP_ID;
```

Output:

EMPLOYEE_DETAILS
198, Donald, Connell, 21-Jun-99, SH_CLERK, 2600, 50
199, Douglas, Grant, 13-Jan-98, SH_CLERK, 3000, 50

200, Jennifer, Whalen, 17-Sep-87, AD_ASST, 4400, 10
201, Michael, Hartstein, 19-Jan-99, IT_PROG, 6000, 20
202, Pat, Fay, 25-Oct-89, AC_MGR, 6500, 20
203, Susan, Mavris, 26-Nov-76, AD_VP, 7500, 40
204, Hermann, Baer, 23-Aug-95, AD_PRES, 9500, 90
205, Shelley, Higgins, 24-Feb-98, AC_MGR, 2300, 60
206, William, Gitz, 12-Mar-01, IT_PROG, 5000, 60
100, Steven, King, 15-Jun-02, AD_ASST, 8956, 100
101, Neena, Kochar, 10-Jul-03, SH_CLERK, 3400, 30

19. Display today's date. Rename the column as TODAY.

SQL Query:

```
SELECT TO_CHAR(SYSDATE,'DD-MON-YYYY') AS TODAY FROM DUAL;
```

Sample output uses the assignment date (6 November 2026); the query returns the actual execution date.

Output:

TODAY
06-NOV-2026

20. Display the employee ID, the day and the year in which the employees were hired.

SQL Query:

```
SELECT EMP_ID, TO_CHAR(HIRE_DATE,'DD') AS HIRE_DAY, TO_CHAR(HIRE_DATE,'YYYY') AS HIRE_YEAR FROM EMP ORDER BY
EMP_ID;
```

Output:

EMP_ID	HIRE_DAY	HIRE_YEAR
198	21	1999
199	13	1998
200	17	1987
201	19	1999
202	25	1989
203	26	1976
204	23	1995
205	24	1998
206	12	2001
100	15	2002
101	10	2003

21. Display the employee name and hire date of all employees in the format 'dd-month-yy'.

SQL Query:

```
SELECT FIRST_NAME || ' ' || LAST_NAME AS EMPLOYEE_NAME, TO_CHAR(HIRE_DATE,'DD-FMMONTH-RR') AS HIRE_DATE FROM EMP
ORDER BY EMP_ID;
```

FM removes Oracle date-format padding. Month names are shown in English.

Output:

EMPLOYEE_NAME	HIRE_DATE
Donald Connell	21-June-99
Douglas Grant	13-January-98
Jennifer Whalen	17-September-87
Michael Hartstein	19-January-99
Pat Fay	25-October-89
Susan Mavris	26-November-76
Hermann Baer	23-August-95
Shelley Higgins	24-February-98
William Gitz	12-March-01

Steven King	15-June-02
Neena Kochar	10-July-03

22. Display the employee ID and the month in which the employees were hired.

SQL Query:

```
SELECT EMP_ID, TO_CHAR(HIRE_DATE,'FMMONTH') AS HIRE_MONTH FROM EMP ORDER BY EMP_ID;
```

Output:

EMP_ID	HIRE_MONTH
198	JUNE
199	JANUARY
200	SEPTEMBER
201	JANUARY
202	OCTOBER
203	NOVEMBER
204	AUGUST
205	FEBRUARY
206	MARCH
100	JUNE
101	JULY

23. Display the employee name, employee ID and hire date in the format 'dd-mon-yyyy'.

SQL Query:

```
SELECT FIRST_NAME || ' ' || LAST_NAME AS EMPLOYEE_NAME, EMP_ID, TO_CHAR(HIRE_DATE,'DD-MON-YYYY') AS HIRE_DATE FROM
EMP ORDER BY EMP_ID;
```

Output:

EMPLOYEE_NAME	EMP_ID	HIRE_DATE
Donald Connell	198	21-Jun-1999
Douglas Grant	199	13-Jan-1998
Jennifer Whalen	200	17-Sep-1987
Michael Hartstein	201	19-Jan-1999
Pat Fay	202	25-Oct-1989
Susan Mavris	203	26-Nov-1976
Hermann Baer	204	23-Aug-1995
Shelley Higgins	205	24-Feb-1998
William Gitz	206	12-Mar-2001
Steven King	100	15-Jun-2002
Neena Kochar	101	10-Jul-2003

24. Display the employee name, employee ID and hire date in the format 'month-dd-yyyy'.

SQL Query:

```
SELECT FIRST_NAME || ' ' || LAST_NAME AS EMPLOYEE_NAME, EMP_ID, TO_CHAR(HIRE_DATE,'FMMONTH-DD-YYYY') AS HIRE_DATE
FROM EMP ORDER BY EMP_ID;
```

Output:

EMPLOYEE_NAME	EMP_ID	HIRE_DATE
Donald Connell	198	JUNE-21-1999
Douglas Grant	199	JANUARY-13-1998
Jennifer Whalen	200	SEPTEMBER-17-1987
Michael Hartstein	201	JANUARY-19-1999
Pat Fay	202	OCTOBER-25-1989
Susan Mavris	203	NOVEMBER-26-1976
Hermann Baer	204	AUGUST-23-1995
Shelley Higgins	205	FEBRUARY-24-1998
William Gitz	206	MARCH-12-2001
Steven King	100	JUNE-15-2002
Neena Kochar	101	JULY-10-2003

25. Display the system year in full spelling (for example, Nineteen Ninety Nine for 1999).

SQL Query:

```
SELECT TO_CHAR(SYSDATE,'YEAR') AS YEAR_IN_WORDS FROM DUAL;
```

Oracle YEAR format spells the current year in English; exact padding/capitalization can depend on NLS settings.

Output:

YEAR_IN_WORDS
TWENTY TWENTY-SIX

26. Find the date 15 days after today's date and 15 days before today's date.

SQL Query:

```
SELECT TO_CHAR(SYSDATE + 15,'DD-MON-YYYY') AS AFTER_15_DAYS, TO_CHAR(SYSDATE - 15,'DD-MON-YYYY') AS BEFORE_15_DAYS
FROM DUAL;
```

Sample output assumes the assignment date is 06-NOV-2026.

Output:

AFTER_15_DAYS	BEFORE_15_DAYS
21-NOV-2026	22-OCT-2026

27. Ensure the domain constraint and entity integrity constraint from the table.

SQL Query:

```
-- These constraints are defined in CREATE TABLE (Question 1).
SELECT CONSTRAINT_NAME, CONSTRAINT_TYPE, SEARCH_CONDITION
FROM USER_CONSTRAINTS
WHERE TABLE_NAME = 'EMP'
ORDER BY CONSTRAINT_NAME;
```

The NOT NULL constraints are also present from Question 1. Verify them using USER_TAB_COLUMNS (NULLABLE = N).

Output:

CONSTRAINT_NAME	CONSTRAINT_TYPE	SEARCH_CONDITION
EMP_DEPT_CK	C	DEPT_ID >= 10
EMP_PK	P	—

28. Drop the domain constraint from the table.

SQL Query:

```
ALTER TABLE EMP DROP CONSTRAINT EMP_DEPT_CK;
```

Output:

```
Output: Table altered.
```

29. Drop the primary key from the table.

SQL Query:

```
ALTER TABLE EMP DROP CONSTRAINT EMP_PK;
```

Output:

```
Output: Table altered.
```

30. Drop the created table.

SQL Query:

```
DROP TABLE EMP;
```

Output:

```
Output: Table dropped.
```